STAT 411-616: Introduction to Bayesian Analysis

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Monte Carlo Sampling

We have seen that Bayesian statisticians use posterior summary statistics to summarize some information from the posterior distribution, for example the posterior mean:

$$E[\theta|x) = \int \theta\, p(\theta|x) d\theta$$

and posterior credible intervals, which are based on quantiles z such that, for example,

$$\int_{-\infty}^{z} p(\theta|x) d\theta = 0.025 \text{ or } \int_{z}^{+\infty} p(\theta|x) d\theta = 0.975,$$

for the 95% credible interval.

Once again, this is very easy to do in the conjugate case: the posterior is of the same distributional form as the prior, with the posterior parameters being simple updates of the prior parameters.

Ordinary Monte Carlo Simulation (OMC)

- The "Monte Carlo method" refers to the theory and practice of learning about probability distributions by simulation rather than calculus.
- In ordinary Monte Carlo (OMC) we use IID simulations from the distribution of interest: suppose $\theta_1, \theta_2, \ldots, \theta_m$ are IID samples from some distribution, and suppose we want to know an expectation

$$\mu = E\{g(\theta)\} = \int g(\theta)p(\theta)d\theta$$

which is the expectation of a function, $g(\cdot)$, of a random variable, θ, having density $p(\cdot)$. The law of large numbers (LLN) says that if we can obtain a sample of m values from p, $\theta_i \overset{i.i.d.}{\sim} p(\cdot)$, then

$$\hat{\mu}_n = \frac{1}{m}\sum_{i=1}^{n} g(\theta_i)$$

converges in probability to μ.

Similarly for the standard deviations,

$$\hat{\sigma}_m = \sqrt{\frac{1}{m}\sum_{i=1}^{m}[g(\theta_i) - \hat{\mu}_n]^2} \xrightarrow{p} sd(\theta),$$

and for the quantiles, etc....

In a OMC:

- the "data" $\theta_1, \ldots, \theta_m$ are computer simulations rather than measurements on objects in the real world,
- the "sample size" is the number of computer simulations rather than the size of some real world data, and
- the unknown parameter μ is in principle completely known, given by some integral, which, unfortunately, we are unable to do.

OMC Terminology

- Often we say that m is the *Monte Carlo sample size* to distinguish it from anything else that may be called a sample size.
- Often we say that $\hat{\mu}_m$ is the *Monte Carlo estimate* or *Monte Carlo approximation* or *Monte Carlo calculation* of μ to distinguish it from anything else that may be called an estimate.
- Often we say that $\hat{\sigma}_m/\sqrt{m}$ is the *Monte Carlo standard error* of $\hat{\mu}_n$ to distinguish it from anything else that may be called a standard error.

Other examples

- Cumulative ordered values approximate $F(\theta)$ (cdf)
- Empirical distribution of the sample $\theta_1, ..., \theta_m$ approximates $p(\theta)$ (use histogram or kernel density estimator)
- $P(g(\theta) > c)$ approximated by proportion of samples where event $(g(\theta^{(i)}) > c)$ occurs
- Sample moments/quantiles/functions approximate true moments/quantiles/functions

Extends easily to higher dimensional parameters.

Normal quantiles

Suppose that $\theta \sim N(0,1)$ and we are interested in

$$Pr(\theta \geq 1) = \int_1^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{\theta^2}{2}} \, d\theta.$$

We cannot compute this integral in closed form

Monte Carlo Integration:

```
n=100000
theta=rnorm(n, 0,1)  ## n samples from a N(0,1)
mean_theta=mean(theta) ## expected value E(\theta)
sd_theta=sd(theta) ## sd(\theta)
Pr_theta_geq_1=mean(theta>=1) ## true value approx 0.16
mean_theta; sd_theta; Pr_theta_geq_1;
```

```
[1] 0.006006067
[1] 0.995561
[1] 0.15784
```

Monte Carlo variance

- If
$$\int \theta^2 p(\theta) d\theta < +\infty$$
the central limit theorem shows that
$$\frac{\bar{\theta}_{MC} - \int \theta p(\theta) d\theta}{\sqrt{\sigma^2/m}} \to N(0,1)$$
where $\sigma^2 = Var[\theta]$. We can obtain the Monte Carlo standard error by means of
$$Var[\theta_{MC}] \simeq \hat{\sigma}^2 = \frac{1}{m} \sum [\theta^{(i)} - \bar{\theta}_{MC}]^2$$
- An approximate 95% Monte Carlo confidence interval for the mean of θ is then $\bar{\theta}_{MC} \pm 1.96\sqrt{\hat{\sigma}^2/m}$, that is we expect the posterior mean of θ to be in this interval for roughly 95% of repeated MC samples.
- To increase accuracy, increase m.

Monte Carlo variance

- Similarly for $g(\theta)$, if

$$\int [g(\theta)]^2 p(\theta) d\theta < +\infty$$

the central limit theorem shows that

$$\frac{\overline{g(\theta)}_{MC} - \int g(\theta)p(\theta)d\theta}{\sqrt{\sigma_g^2/m}} \to N(0,1)$$

where $\sigma_g^2 = Var[g(\theta)]$. We can obtain confidence bounds for $\overline{g(\theta)}_{MC}$ and estimate the asymptotic variance by means of

$$Var[g(\theta)_{MC}] \simeq \hat{\sigma}_g^2 = \frac{1}{m}\sum [g(\theta^{(i)}) - \overline{g(\theta)}_{MC}]^2$$

- An approximate 95% Monte Carlo confidence interval for the mean of $g(\theta)$ is $\overline{g(\theta)}_{MC} \pm 1.96\sqrt{\hat{\sigma}_g^2/m}$.

Choice of Prior Distribution

What is the interpretation of probability?

- Objective Probability
  As normative and objective representations of what is rational to believe about a parameter, usually in a state of ignorance (dice, coin, etc.)
- Frequentist probability
  Long run frequency when a process is repeated
- Subjective Probability
  Personal judgment about an event. But, one must be "coherent".

Two general types of Priors:

- Subjective Priors
  Priors chosen to reflect expert opinion or personal beliefs
- Objective Priors
  Priors chosen to let the data (i.e., likelihood) dominate the posterior distribution, and hence inference. These are generally determined based on the sampling model in use.

Subjective Priors

Ways of Specifying

- For single events:
  Relative probability of A to "not A" (as for lotteries/betting)
- Continuous quantities: divide range into intervals and assign probabilities for each (as for events above) or specify percentiles (use *findBeta()* function in the LearnBayes R package).
- Examples: (i) $\theta \sim Beta(\alpha, \beta)$. From expert: mean=.3 and s.d.=0.1. Then solve for α and β and find Beta(9.2,13.8). (ii) $\theta \sim N(\mu, s)$. From expert: an educated guess of 25 and a 95% confidence that θ is between 10 and 40. Then, $\theta \sim N(25, 30/4)$.

Subjective priors may be classified into two classes:

- Conjugate Priors
- Non-conjugate priors

"Objective" Priors

Goal: Choose priors reflecting a state of "no-information" or "ignorance" about θ.

For example, if θ can only have a finite set of values, it seems natural to assume all values equally likely a priori. Such specifications fall into the general class of noninformative priors. Roughly speaking a prior is non-informative if it has minimal impact on the posterior, i.e. it does not change much over the region where the likelihood is appreciable.

Default/Flat Priors:
1) $x \sim Bin(n, \theta)$, $\theta \sim Beta(1, 1) \equiv U(0, 1)$
2) $x \sim N(\theta, \sigma^2)$, θ unknown, σ known. Then $\pi(\theta) = 1$ for $-\infty < \theta < \infty$
3) $X \sim Poisson(\lambda)$. Then $\pi(\lambda) \sim 1$ for $0 < \lambda < \infty$ (also $Ga(\alpha, \beta \to 0)$).

Pros/Cons of these priors:

- Non-informative: All values of θ are equally likely. No information.
- Likelihood dominates posterior: $p(\theta|x) \propto L(\theta)\pi(\theta) = L(\theta)$
- May not be a probability distribution, i.e. $\int \pi(\theta)d\theta = \infty$ ("improper").

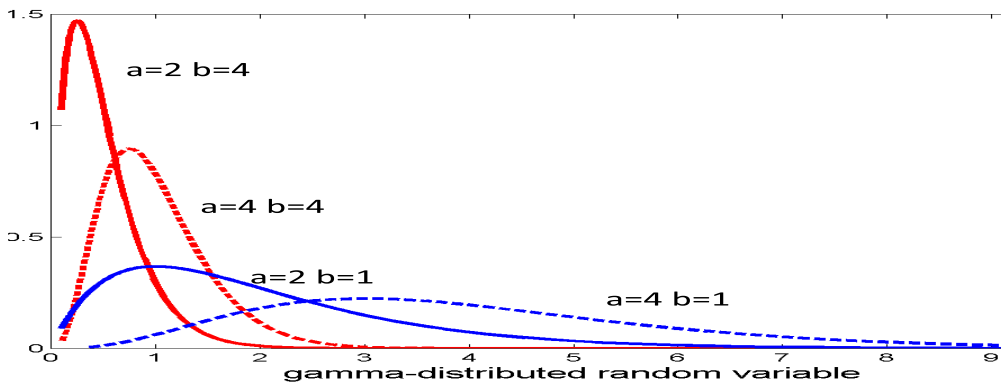
Diffuse Priors

Motivation is to use a proper prior with "little" info about the parameter.

- Conjugate priors are proper
- The spread of a prior is a measure of the amount of uncertainty about the parameter expressed by the prior.

So, choose a conjugate prior with large standard deviation as a *diffuse* or *weakly informative* prior.

Examples: (i) $X \sim Poisson(\lambda), \;\; 0 < \lambda < \infty, \;\; \lambda \sim Ga(\alpha, \beta)$. A diffuse prior is $Ga(\alpha, 1)$ with α the prior expectation.



(ii) $X \sim N(\theta, \sigma^2)$ with σ^2 known. A diffuse prior for θ is $N(\mu_0, 1000)$.

Multi-parameter models

- The Normal model
- The multinomial (exercise)

Parameterizations of the Normal Distribution

- Mean and deviation:

$$f(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R}, \sigma > 0.$$

- Mean and precision:

$$f(x|\mu,\tau) = \sqrt{\frac{\tau}{2\pi}} e^{-\frac{\tau(x-\mu)^2}{2}}, x \in \mathbb{R}, \tau = \frac{1}{\sigma^2} > 0.$$

- The latter has advantages in numerical computations when $\sigma \to 0$ and simplify formulas.

Summary

	pdf/pmf	Domain	Mean	Variance
Bern	$P(x) = p^x(1-p)^{1-x}$	$\{0,1\}$	p	$p(1-p)$
Bin	$P(x) = \begin{pmatrix} N \\ x \end{pmatrix} p^x(1-p)^{N-x}$	$\{0,\ldots,N\}$	Np	$Np(1-p)$
Poi	$P(x) = e^{-\lambda}\frac{\lambda^x}{x!}$	$\mathbb{N}$	λ	λ
NB	$P(x) = \begin{pmatrix} r+x-1 \\ x \end{pmatrix} p^r(1-p)^x$	$\mathbb{N}$	$r\frac{1-p}{p}$	$r\frac{1-p}{p^2}$
M	$P(x_1,\ldots,x_k) = \frac{N!}{\prod_k x_k!}\prod_k p_k^{x_k}$	$\{0,\ldots,N\}^K$	$N\mathbf{p}$	$\begin{cases} Np_k(1-p_k) \\ -Np_kp_k' \end{cases}$
U	$f(x) = \frac{1}{b-a}$	$[a,b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Be	$f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}x^{a-1}(1-x)^{b-1}$	$[0,1]$	$\frac{a}{a+b}$	$\frac{ab}{(a+b)^2(a+b+1)}$
Ga	$f(x) = \frac{b^a}{\Gamma(a)}x^{a-1}e^{-bx}$	$\mathbb{R}^+$	$\frac{a}{b}$	$\frac{a}{b^2}$
N	$f(x) = \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\mathbb{R}$	μ	σ^2
MN	$f(X) = (2\pi)^{-\frac{p}{2}}\lvert\Sigma\rvert^{-\frac{1}{2}} e^{-\frac{1}{2}(X-\mu)^T\Sigma^{-1}(X-\mu)}$	$\mathbb{R}^p$	μ	Σ

Summary

	Model parameters	MOM	MLE	UMVUE
Bern	p	$\bar{X}$	$\bar{X}$	$\bar{X}$
Bin	p	$\frac{\bar{X}-\frac{n-1}{n}S^2}{\bar{X}}$	$\frac{\bar{X}}{nN}$	$\frac{\bar{X}}{nN}$
Poi	λ	$\bar{X}$	$\bar{X}$	$\bar{X}$
NB	r	$\frac{\bar{X}^2}{\bar{X}-\frac{n-1}{n}S^2}$	with known r	
	p	$\frac{\hat{r}}{\hat{r}+\bar{X}}$	$\frac{r}{r+\bar{X}}$	
U	a	$\bar{X}-\sqrt{3\frac{n-1}{n}S^2}$	$X_{(1)}$	with $a=0$
	b	$\bar{X}+\sqrt{3\frac{n-1}{n}S^2}$	$X_{(n)}$	$\frac{n+1}{n}X_{(n)}$
Ga	a	$\frac{\bar{X}^2}{\frac{n-1}{n}S^2}$	with known a	
	b	$\frac{\bar{X}}{\frac{n-1}{n}S^2}$	$\frac{a}{\bar{X}}$	
N	μ	$\bar{X}$	$\bar{X}$	$\bar{X}$
	σ^2	$\frac{n-1}{n}S^2$	$\frac{n-1}{n}S^2$	with known σ^2

Possible models

- Data likelihood:

$$
\begin{aligned}
f(x_1, \ldots, x_n | \mu, \sigma^2) &= \prod_{i=1}^{n} f(x_i | \mu, \sigma^2) \\
&= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \\
&= \left(2\pi\sigma^2\right)^{-\frac{n}{2}} e^{-\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}}.
\end{aligned}
$$

- Models:
  - μ is unknown, σ^2 is known;
  - μ is known, σ^2 is unknown;
  - Both μ and σ^2 are unknown:
    - μ is dependent on σ^2;
    - μ and σ^2 are independent.

(see addendum for complete cases)

The Normal Model

$x = (x_1, \ldots, x_n) \sim N(\mu, \sigma^2)$ i.i.d., with both μ and σ unknown. The likelihood is:

$$\begin{aligned} L(\mu, \sigma^2) &\propto \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{1}{2\sigma^2}(x_i - \mu)^2 \Big) \\ &\propto \Big(\frac{1}{\sigma^2}\Big)^{n/2} \exp\Big(-\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2 \Big) \end{aligned}$$

For inference, focus is on $p(\mu, \sigma^2|x) = p(\mu|\sigma^2, x)p(\sigma^2|x)$.

From a Bayesian perspective, it is easier to work with the precision, $\tau = \frac{1}{\sigma^2}$. The likelihood becomes:

$$\begin{aligned} L(\mu, \tau) &\propto \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} \tau^{1/2} \exp\Big(-\frac{1}{2}\tau(x_i - \mu)^2 \Big) \\ &\propto \tau^{n/2} \exp\Big(-\frac{1}{2}\tau \sum_i (x_i - \mu)^2 \Big) \end{aligned}$$

Likelihood factorization:

$$
\begin{aligned}
L(\mu,\tau) &\propto \tau^{n/2}\exp\Big(-\frac{1}{2}\tau\sum_i(x_i-\mu)^2\Big)\\
&\propto \tau^{n/2}\exp\Big(-\frac{1}{2}\tau\sum_i[(x_i-\bar{x})-(\mu-\bar{x})]^2\Big)\\
&\propto \tau^{n/2}\exp\Big(-\frac{1}{2}\tau\Big[\sum_i(x_i-\bar{x})^2+n(\mu-\bar{x})^2\Big]\Big)\\
&\propto \tau^{n/2}\exp\Big(-\frac{1}{2}\tau s^2(n-1)\Big)\exp\Big(-\frac{1}{2}\tau n(\mu-\bar{x})^2\Big)\\
&\propto \tau^{n/2}\exp\Big(-\frac{1}{2}\tau SS\Big)\exp\Big(-\frac{1}{2}\tau n(\mu-\bar{x})^2\Big)
\end{aligned}
$$

with $s^2=\sum_i(x_i-\bar{x})^2/(n-1)$ and $SS=\sum_i(x_i-\bar{x})^2$ sample variance and sum of squares [SS and $\bar{x}$ sufficient statistics]

Non-informative Prior

Non-informative prior: $\pi(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$. This arises by considering μ and σ^2 a priori independent and taking the product of the standard non-inf priors. This is not a conjugate setting (the posterior does not factor into a product of two independent distributions). Prior is improper but posterior is proper. This is also the Jeffreys' prior.

Joint posterior distribution of μ and σ^2 is

$$p(\mu, \sigma^2|x) \propto (\sigma^2)^{-(n/2+1)} \exp\left\{-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{x}-\mu)^2]\right\}$$

where

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- The conditional posterior distribution, $p(\mu|\sigma^2, x)$, is equivalent to deriving the posterior for μ when σ^2 is known

$$\mu|\sigma^2, x \sim \mathcal{N}\left(\bar{x}, \frac{\sigma^2}{n}\right)$$

- The marginal posterior $p(\sigma^2|x)$, is obtained integrating $p(\mu, \sigma^2|x)$ over μ [Hint: integral of a Gaussian function $2c\sqrt{\pi} = \int 2\exp(-\frac{1}{c^2}(\mu+b)^2)d\mu$]

$$\begin{aligned} p(\sigma^2|x) &\propto \int_\mu (\sigma^2)^{-(n/2+1)} \exp\left\{-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{x}-\mu)^2]\right\} d\mu \\ &\propto (\sigma^2)^{-[(n-1)/2+1]} \exp\left\{-\frac{(n-1)s^2}{2\sigma^2}\right\} \end{aligned}$$

which is an inverse-gamma density, i.e.

$$\sigma^2|x \sim \text{Inv-Gamma}\left(\frac{n-1}{2}, \frac{n-1}{2}s^2\right) \equiv \text{Inv-}\chi^2(n-1, s^2)$$

or, equivalently, $\tau|x \sim Ga$.

Sampling from the joint posterior distribution

One can simulate a value of (μ, σ^2) from the joint posterior density by

1. simulating σ^2 from an inverse-Gamma$\left(\frac{n-1}{2}, s^2\frac{n-1}{2}\right)$ distribution [take the inverse of random samples from a Gamma$\left(\frac{n-1}{2}, s^2\frac{n-1}{2}\right)$]
2. then simulating μ from $\mathcal{N}\left(\bar{x}, \frac{\sigma^2}{n}\right)$ distribution.

Marginal posterior distribution $p(\mu|x)$ of μ

As μ is typically the parameter of interest (σ^2 nuisance parameter) it is useful to calculate its marginal posterior distribution

[Hint: integral of a Gamma function $\frac{\Gamma(a)2^a}{b^a} = \int_0^\infty z^{a-1}\exp(-\frac{zb}{2})dz$]

$$
\begin{aligned}
p(\mu|x) &= \int_0^\infty p(\mu,\sigma^2|x)d\sigma^2 \\
&\propto \int_0^\infty (\sigma^2)^{-(n/2+1)}\exp\left\{-\frac{1}{2\sigma^2}[(n-1)s^2+n(\bar{x}-\mu)^2]\right\}d\sigma^2 \\
&= A^{-n/2}\int_0^\infty z^{(n-2)/2}\exp(-z)dz, \\
&\quad \text{with } A=(n-1)s^2+n(\bar{x}-\mu)^2, z=\frac{A}{2\sigma^2} \\
&\propto A^{-n/2} = \left[1+\frac{1}{n-1}\left(\frac{\mu-\bar{x}}{s/\sqrt{n}}\right)^2\right]^{-[(n-1)+1]/2}
\end{aligned}
$$

that is, $\mu|x \sim t(n-1,\bar{x},s^2/n)$, or $\frac{\mu-\bar{x}}{s/\sqrt{n}}|x \sim t_{n-1}$ with t_{n-1} the standard t-distribution with $n-1$ degrees of freedom

Conjugate Prior Model

A conjugate prior must be of the form $\pi(\mu, \sigma^2) = \pi(\mu|\sigma^2)\pi(\sigma^2)$, e.g.,

$$\mu|\sigma^2 \sim N(\mu_0, \sigma^2/\tau_0), \quad \sigma^2 \sim IG\Big(\frac{\nu_0}{2}, \frac{SS_0^2}{2}\Big) \ \Big[\text{or } \tau \sim Ga\Big],$$

which corresponds to the joint prior density

$$p(\mu, \sigma^2) \propto \left(\frac{\sigma^2}{\tau_0}\right)^{-1/2} \exp\left\{-\frac{1}{2\sigma^2/\tau_0}(\mu - \mu_0)^2\right\} (\sigma^2)^{-(\nu_0/2+1)} \exp\left\{-\frac{SS_0^2}{2\sigma^2}\right\}$$
$$= (\sigma^2)^{-\left(\frac{\nu_0+1}{2}+1\right)} \exp\left\{-\frac{\tau_0}{2\sigma^2}\left(\frac{SS_0^2}{\tau_0} + (\mu - \mu_0)^2\right)\right\}$$

we call this a Normal-Inverse-Gamma prior,

$$(\mu, \sigma^2) \sim NIG(\mu_0, \tau_0, \nu_0/2, SS_0/2)$$

Joint Posterior $p(\mu, \sigma^2|y)$

$$
\begin{aligned}
p(\mu, \sigma^2|x) &\propto (\sigma^2)^{-\left(\frac{\nu_0+1}{2}+1\right)} \exp\left\{-\frac{1}{2\sigma^2}\left(SS_0^2 + \tau_0(\mu-\mu_0)^2\right)\right\} \\
&\quad \times (\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2\right\} \\
&\propto (\sigma^2)^{-\left(\frac{\nu_n+1}{2}+1\right)} \exp\left\{-\frac{\tau_n}{2\sigma^2}\left(\frac{SS_n^2}{\tau_n} + (\mu-\mu_n)^2\right)\right\}
\end{aligned}
$$

with

$$
\mu|\sigma^2, x \sim N(\mu_n, \sigma^2/\tau_n), \quad \mu_n = \frac{\mu_0\frac{\tau_0}{\sigma^2} + \bar{x}\frac{n}{\sigma^2}}{\frac{\tau_0}{\sigma^2} + \frac{n}{\sigma^2}} = \frac{\tau_0\mu_0 + n\bar{x}}{\tau_n}, \ \tau_n = \tau_0 + n
$$

$$
\sigma^2|x \sim IG\left(\frac{\nu_n}{2}, \frac{SS_n^2}{2}\right), \quad \nu_n = \nu_0 + n, SS_n^2 = SS_0^2 + SS + \frac{\tau_0 n}{\tau_n}(\bar{x}-\mu_0)^2
$$

Thus, $\mu, \sigma^2|y \sim$ Normal-Inverse Gamma$(\mu_n, \tau_n; \nu_n/2, SS_n^2/2)$.

$$\text{Also } \mu|x \quad \sim \quad t_{\nu_n}(\mu_n, \sigma_n^2/\tau_n), \quad \sigma_n^2 = SS_n^2/\nu_n$$

[Note: Again $\int N(m, \sigma^2/\tau) IG(\nu/2, SS/2) d\sigma^2 = t_\nu(m, SS/(\nu\tau)]$

Comments:

- μ_n expected value for μ after seeing the data

$$\mu_n = \frac{n}{\tau_n}\bar{x} + \frac{\tau_0}{\tau_n}\mu_0, \quad \text{weighted average}$$

- τ_n precision for estimating μ after n observations.
- ν_n degrees of freedom
  $[\tau \sim Ga(\alpha/2, \beta/2) \to \beta\tau \sim \chi^2_\alpha, \text{ with } \alpha \text{ degrees of freedom}]$
- SS_n posterior variation as prior variation+observed variation+variation between prior mean and sample mean.
- Limiting case $\tau_0 \to 0$, $\nu_0 \to -1$ (and $SS \to 0$) then $\mu|x \sim t_{n-1}(\bar{x}, s^2/n)$ (same as improper prior!)

Example on SPF (from Merlise Clyde)

A Sunlight Protection Factor (SPF) of 5 means an individual that can tolerate X minutes of sunlight without any sunscreen can tolerate $5X$ minutes with sunscreen. Data on 13 individual (tolerance, in min, with and without sunscreen).

Analysis should take into account pairing which induces dependence between observations (take differences and use ratios or log(ratios) = difference in logs). Ratios make more sense given the goals: how much longer can a person be exposed to the sun relative to their baseline.

Model: $Y = \log(TRT) - \log(CONTROL) \sim N(\mu, \tau)$. Then $E(\log(TRT/CONTROL)) = \mu = \log(SPF)$. Interested in $\exp(\mu) = SPF$. Summary statistics: $\bar{y} = 1.998, s^2 = 0.525, n = 13$

[make boxplots and Q-Q normal plots to check on normality]

- Model formulation: $Y = \log(\text{TRT}) - \log(\text{CONTROL}) \sim \text{N}(\mu, \sigma^2)$, $n = 13$, $\bar{y} = 1.998$, $SS = 0.525$.
- Question: $\pi(\mu|y_1, \ldots, y_n) = ?$
- Bayesian model:
  - Data likelihood: $f(y_1, \ldots, y_n|\mu, \sigma^2) = \prod_{i=1}^{n} \text{N}(y_i; \mu, \sigma^2)$;
  - Non-informative Prior: $(\mu, \sigma^2) \sim 1/\sigma^2$;
  - Posterior: $(\mu, \sigma^2|y_1, \ldots, y_n) \sim N(\bar{y}, \sigma^2/n)IG(\frac{n-1}{2}, s^2\frac{n-1}{2})$
  - Posterior: $\mu|y_1, \ldots, y_n \sim t_{n-1}(\bar{y}, \frac{s^2}{n})$;
  - Prediction: $y_f|y_1, \ldots, y_n \sim \text{t}_{n-1}(\bar{y}, s^2(n-1)/n)$.
- Coding in R: rgamma(), rnorm() and rt().

With non-informative prior.

Posterior: $(\mu, \sigma^2 | y_1, \ldots, y_n) \sim N(\bar{y}, \sigma^2/n) IG(\frac{n-1}{2}, s^2 \frac{n-1}{2})$

Posterior: $\mu | y_1, \ldots, y_n \sim t_{n-1}(\bar{y}, \frac{s^2}{n})$

Define: $vn = (n-1) = 12, SSn = s^2(n-1) = 0.525, mn = 1.998$

Sampling from posterior:
Draw $\tau | Y$

```
tau = rgamma(10000, vn/2, rate=SSn/2)
```

Draw $\mu | \tau, Y$

```
mu = rnorm(10000, mn, 1/sqrt(phi*n))
```

or draw $\mu | Y$ directly

```
mu = rt(10000,vn)*sqrt(SSn/(n*vn))+ mn
```

- Model formulation: $Y = \log(\text{TRT}) - \log(\text{CONTROL}) \sim \text{N}(\mu, \sigma^2)$, $n = 13$, $\bar{y} = 1.998$, $SS = 0.525$.
- Question: $\pi(\mu|y_1, \ldots, y_n) = ?$
- Bayesian model:
  - Data likelihood: $f(y_1, \ldots, y_n|\mu, \sigma^2) = \prod_{i=1}^{n} \text{N}(y_i; \mu, \sigma^2)$;
  - Conjugate Prior: $\mu|\sigma^2 \sim \text{N}(\mu_0, \frac{\sigma^2}{\tau_0})$, $\sigma^2 \sim \text{IG}(\frac{\nu_0}{2}, \frac{SS_0}{2})$;
  - Posterior: $(\mu, \sigma^2|y_1, \ldots, y_n) \sim NIG(\mu_n, \tau_n; \nu_n/2, SS_n^2)$
  - Posterior: $\mu|y_1, \ldots, y_n \sim t_{\nu_n}(\mu_n, \frac{SS_n}{\tau_n \nu_n})$;
  - Prediction: $y_f|y_1, \ldots, y_n \sim \text{t}_{\nu_n}(\mu_n, \frac{SS_n}{\nu_n} \frac{\tau_n+1}{\tau_n})$.
- Coding in R: rgamma(), rnorm() and rt().

Expert opinions on μ:
· Best guess on median SPF is 16
· $P(\mu > 64) = 0.01$
· information in prior is worth 25 observations

Possible subjective prior: $\mu_0 = log(16)$, $\tau_0 = 25$, $\nu_0 = \tau_0 - 1$
$P(\mu < \log(64)) = .99$ implies $SS_0 = 185.7$

Posterior hyperpar: $\tau_n = 38, \mu_n = 2.508, \nu_u = 37, SS_n = 197.134$

Sampling from posterior:
Draw $\tau|Y$

```
tau = rgamma(10000, vn/2, rate=SSn/2)
```

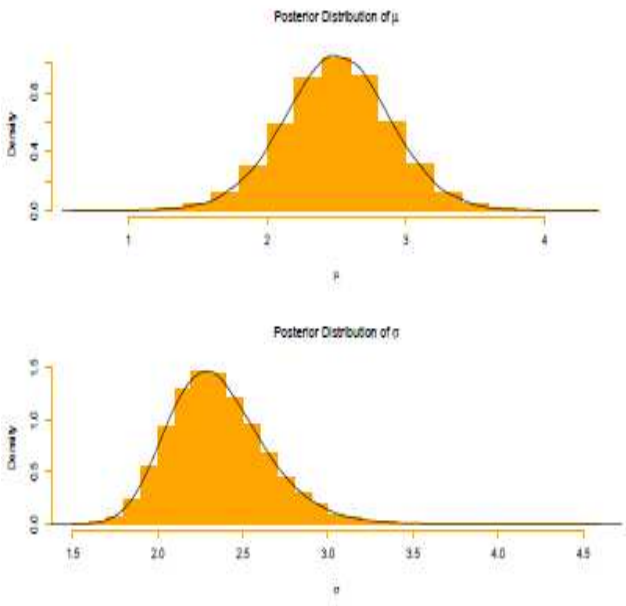
Draw $\mu|\tau, Y$

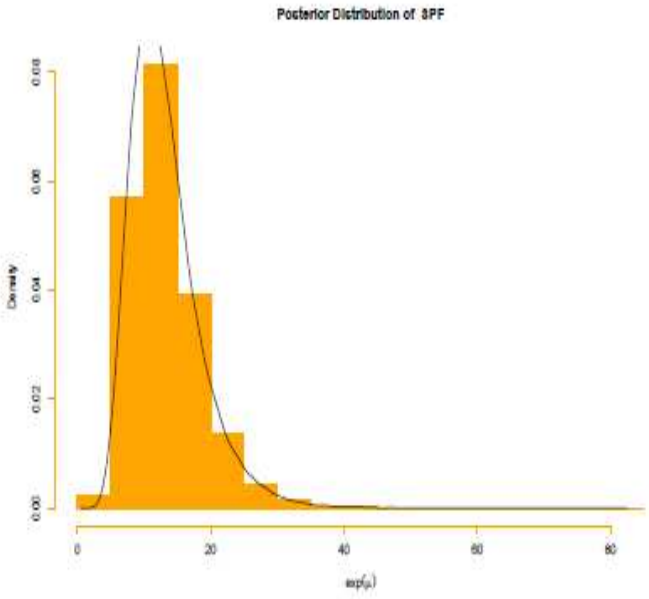
```
mu = rnorm(10000, mn, 1/sqrt(phi*tn))
```

or draw $\mu|Y$ directly

```
mu = rt(10000,vn)*sqrt(SSn/(tn*vn))+ mn
```

Transform to $\exp(\mu)$. Find 95% C.I. of 4.54 to 23.758





Predictive Distribution of future z

- Posterior predictive distribution (given $x = (x_1, \ldots, x_n)$):

$$p(z|x) = \int \int p(z|\mu, \sigma^2, x) p(\mu, \sigma^2|x) d\mu d\sigma^2$$

[Use assumption that z is independent of x given μ and σ^2, then integrate μ using the normal integral, then integrate σ^2 using the Gamma integral]

· Reference prior: $z|x \sim t_{n-1}\left(\bar{x}, s^2(n+1)/n\right)$

· Conjugate prior: $z|x \sim t_{\nu_n}\left(\mu_n, \sigma_n^2(\tau_n + 1)/\tau_n\right), \quad \sigma_n^2 = SS_n^2/\nu_n$

Alternatively, MC sampling: sample from joint posterior $p(\mu, \sigma^2|x)$ and then from $p(z|\mu^{(i)}, \sigma^{(i)}, x)$

Back to example

Prior predictive distribution: $z \sim t_{24}\Big(\log(16), \frac{185.7}{24}(1+\frac{1}{25})\Big)$

Posterior predictive distribution: $z \sim t_{37}\Big(2.5, 5.32(1+\frac{1}{38})\Big)$

```
Y=rt(10000,24)*sqrt((1+1/25)*187.5/24)+log(16)
quantile(exp(Y))
0% 25% 50% 75% 100%
4.57e-06 2.32 16.78 114.98 370966.2
```

Sampling from posterior predictive leads to 50% C.I. (0.0003,12.4) - with sunscreen, 50% chance that next individual can be exposed from 0 to 12 times longer than without sunscreen.

Semi-conjugate prior

A semi-conjugate setting is obtained with independent priors
$\pi(\mu, \sigma^2) = \pi(\mu)\pi(\sigma^2)$

$$\mu \sim N(\mu_0, \sigma_0^2), \quad \sigma^2 \sim IG\Big(\frac{\nu_0}{2}, \frac{SS_0}{2}\Big)$$

$$\text{then} \quad \mu|\sigma^2, x \quad \sim \quad N(\mu_n, \tau_n^2), \quad \mu_n = \frac{\frac{\mu_0}{\sigma_0^2} + \bar{x}\frac{n}{\sigma^2}}{\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}}, \quad \tau_n^2 = \frac{1}{\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}}$$

$$\sigma^2|x \quad \sim \quad \text{not in closed form}$$

We will solve this with MCMC methods!

Proper (semi-conjugate) reference (diffuse) priors

- More recently priors such as

$$\mu \sim N(0, b)$$
$$\tau \sim Gamma(c, c)$$

  have been used as proper reference priors.
- In this case, b and c are chosen so that the prior precision for μ, $1/b$, and both hyperparameters c in the Gamma distribution are near zero.
- Such priors are seen as approximation of the $p(\mu, \tau) \propto 1/\tau$ improper default prior.
- Common choices are $b = 10^6$ and $c = 0.001$.

Back to the example

- We need to identify a prior distribution that gives information/no-information about the unknown parameters μ and $\tau = 1/\sigma^2$.
- $\mu \sim N(0, 10^6)$ as proper non-informative prior.
- Expert opinion that μ should be centered at 16. Then, $\mu \sim N(16, 10^6)$ as diffuse prior.
- Expert 95% certain that the mean SPF should be μ should be between 10 and 75, that is, $Pr(10 < \mu < 75) = 0.95$. Then $\mu \sim N(16, 0.0163)$
- We have no good information on σ^2, the variance of an observation
- So we can specify a reference (vague) prior on τ, which is independent of μ:

$$\tau \sim Gamma(0.001, 0.001)$$

The Multinomial Model

- $x = (x_1, \ldots, x_k)$ vector of counts, with x_j =number of observations for the $j - th$ category and $\sum x_j = n$
- $p(x|\theta) \propto \prod_{j=1}^{k} \theta_j^{x_j}$ with $\sum_j \theta_j = 1$
- Conjugate prior is Dirichlet, $\theta \sim D(\alpha)$

$$\pi(\theta) \propto \prod_{j=1}^{k} \theta_j^{\alpha_j - 1}$$

$$p(\theta|x) \propto \prod_{j=1}^{k} \theta_j^{x_j + \alpha_j - 1}$$

that is, $\theta|x \sim D(\alpha + x)$

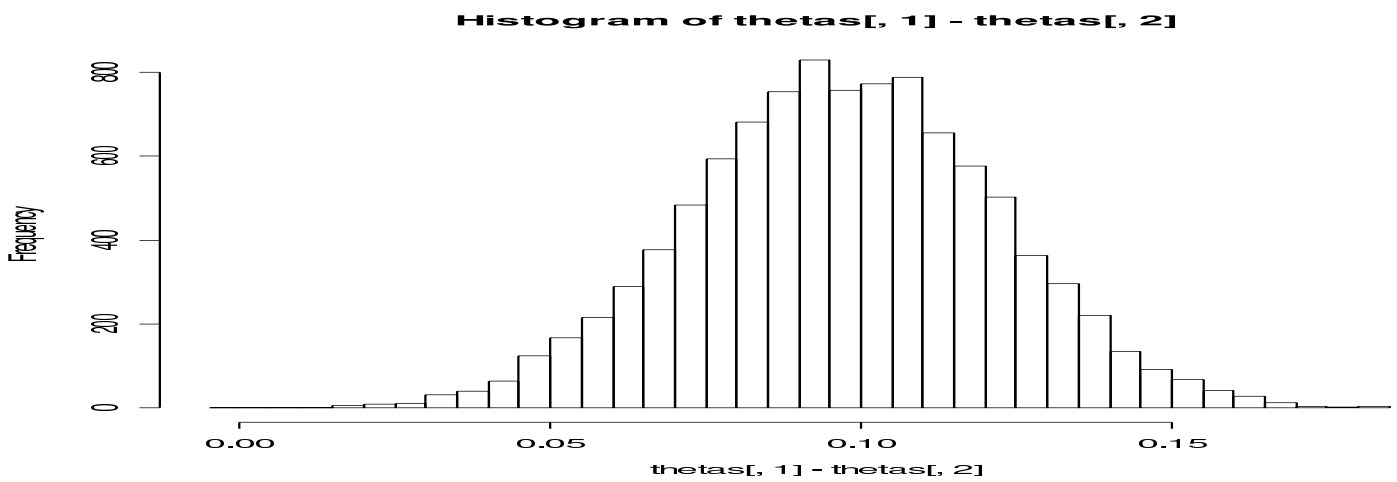
- Uniform prior if $\alpha_j = 1$ for every j. Improper prior if $\alpha_j = 0$ for every j (uniform on $\log \theta_j$) but with proper posterior if there is at least one obs in each of the k categories (i.e., $x_j > 0$ for each j).

Example

Survey of 1447 US voters to find out their preferences in the upcoming presidential election. Data: $x_1 = 727$ support the republican candidate, $x_2 = 583$ the democratic candidate, $x_3 = 173$ have no preference.

$$p(x|\theta) \propto \theta_1^{727}\theta_2^{583}\theta_3^{173}, \;\; \theta \sim D(1,1,1), \;\; \theta|x \sim D(728, 584, 174)$$

Histogram of $\theta_1 - \theta_2$ indicates more support for republican candidate




```
thetas <- rdirichlet(10000,c(728,584,174))
hist(thetas[,1]-thetas[,2], nclass=50)
```

Prelude to MCMC methods

Often the posterior distribution of a parameter θ or a function of it, $g(\theta)$, cannot be derived in closed form.
Example: Non-conjugate priors. For the normal model $x \sim N(\mu, \sigma^2)$, the independent prior $\pi(\mu, \sigma^2) = \pi(\mu)\pi(\sigma^2)$ with $\mu \sim N(\mu_0, \sigma_0^2)$ and $\sigma^2 \sim 1/\sigma^2$ (or Inv-Ga) leads to $p(\sigma^2|x)$ which is not of a familiar form.

Need alternative methods for computing posterior distributions and post summaries. Available options:

- Analytical methods based on approximations, e.g., large sample normal approximation of the posterior distribution, Laplace approximation methods, numerical integration (will not be covered).
- Simulation methods bases on direct sampling from the posterior (rejection sampling, importance sampling).
- Simulate from a Markov chain whose stationary distribution is the desired posterior distribution (e.g., via Gibbs sampler and Metropolis-Hastings algorithms) and then calculate MC estimates.